MRPFA-Net for Shadow Detection in Remote-Sensing Images

Jing Zhang, Xinlong Shi, Congyao Zheng, Jun Wu, and Yunsong Li, *Member, IEEE*

***Abstract*— The presence of shadows in high-resolution (HR) remote-sensing images reduces object detection accuracy. To address this problem, in this article, we proposed a deep neural network algorithm for shadow detection by using the aerial imagery dataset for shadow detection (AISD) and shadow semantic annotation database (SSAD) remote-sensing shadow image datasets. To improve the ability to extract spatial information from feature maps, we developed a cross-spatial attention module that focuses on semantic information in the horizontal and vertical directions at each position point on the remote-sensing image. This module overcomes the limitations of existing technologies in accurately judging small areas and suspected shadow areas and in missing or incorrectly detected shadow areas. In addition, to improve the ability to extract shadow features and the accuracy of shadow detection in remote-sensing images, we developed a channel attention module that assigns more attention to channels that conform to the shadow color characteristics. The network architecture comprises an encoder–decoder structure, with ResNeXt50 used as the backbone for the encoder and a multiresolution parallel fusion (MRPF) designed for the decoder; cross-spatial and channel attention were incorporated into the decoder unit. Experimental results demonstrated the superior performance of the proposed algorithm, with an F1 score of 92.6% for the shadow category on the test set, thus outperforming other algorithms and making the proposed method an effective solution for shadow detection in HR remote-sensing images.**

***Index Terms*— Convolutional neural network (CNN), deep learning (DL), remote sensing, shadow detection.**

I. Introduction

HIGH-RESOLUTION (HR) remote-sensing images can capture complex details regarding land cover, such as building structures, road conditions, and vegetation textures, and are increasingly used in practical applications, including surface feature information detection. However, tall ground objects obstructing oblique rays cause shadows in remote-sensing images. These shadows cause inconvenience in various computer vision tasks, such as the detection, segmentation, and tracking of ground objects. Ground object features in shadows cannot be completely extracted, and local features get damaged, affecting the accuracy of ground object detection [1], [2]. Therefore, automatic detection and removal of shadows are crucial for facilitating ground object information detection. Beyond this, in cartography, shadow detection helps generate accurate maps by resolving ambiguities; in environmental science, it accurately tracks land cover changes; in urban planning, it better quantifies built structures; and in defense, it reveals hidden information. In addition, shadowless images are valuable for image editing, computational photography, and augmented reality. Thus, it is essential to process remote-sensing image shadows effectively.

Shadow processing involves two processes: shadow detection and shadow removal. Shadow detection is performed to determine the spatial position of shadows in the image, laying the foundation for shadow processing. Shadow removal is performed to recover the distorted radiation information in the shadow region. The quality of shadow removal mainly depends on the accuracy of shadow detection. Therefore, good shadow detection is necessary to achieve better shadow removal.

In this study, we aimed to explore shadow detection and develop an automatic and high-accuracy shadow detection method based on deep learning (DL), laying the foundation for restoring the information of objects under shadow occlusion.

Existing shadow detection methods can be classified as model-based and shadow property-based methods.

The model-based method [3], [4], [5] identifies shadows by constructing physical or geometric models and requires prior knowledge of scenes and sensors, such as scene distribution, terrain patterns, weather conditions, sensor parameters, and acquisition directions. This method yields high detection accuracy but relies heavily on prior information and may fail when limited prior information is available. This method is gradually losing popularity in the research field.

The shadow property-based method [6], [7], [8] uses some specific shadow attributes in the image (e.g., brightness, chroma, spectrum, and geometric features) and designs shadow features manually. It is divided into the following four submethods according to the algorithm used to distinguish shadows:

1) *Threshold Segmentation Method:* This method divides the image into shadow and nonshadow regions by threshold according to the difference in the visual

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Jing Zhang is with the State Key Laboratory of Integrated Services Networks and the School of Telecommunication Engineering, Xidian University, Xi'an 710071, China, also with the Hangzhou Institute of Technology, Xidian University, Hangzhou 311231, China, and also with the Guangzhou Institute of Technology, Xidian University, Guangzhou 510555, China (e-mail: jingzhang@xidian.edu.cn).

Xinlong Shi, Congyao Zheng, and Jun Wu are with the School of Telecommunication Engineering, Xidian University, Xi'an 710071, China.

Yunsong Li is with the State Key Laboratory of Integrated Services Networks and the School of Telecommunication Engineering, Xidian University, Xi'an 710071, China.

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characteristics between shadow and nonshadow areas in terms of brightness, saturation, and so on [9], [10], [11]. This method includes two main steps. First, the image is converted into a new feature space in which the shadow and nonshadow areas can be easily separated. Next, binarization is performed using the threshold obtained from the bimodal distribution of the image histogram so that the shaded area and the unshaded area can be distinguished. The accuracy of this method depends only on the segmentation threshold. However, remote-sensing images often have complex scenes and diverse surface types, and a single segmentation threshold cannot be applied to a variety of shadow situations.

2) *Soft Shadow Detection Method:* This method generates a shadow probability map through image matting to describe the probability that each pixel belongs to shadow [12], [13], [14], that is, soft shadow. Soft shadows can depict shadows more clearly than hard shadows and aid in solving the problem of uneven shadows. However, this method requires marking the shadow and nonshadow areas on each image in advance, which is time-consuming, labor-intensive, and subjective. In addition, the shadow probability map needs to select the appropriate threshold for segmentation according to the specific image scene, and its generalization ability is poor.
3) *Traditional Machine-Learning (ML) Method:* This method uses perceptron [15], [16], support vector machines [12], [17], and other methods to detect shadows. In simple scenes, the perceptron and support vector machines can achieve high detection accuracy; however, when in complex scenes, the manually designed shadow features change with the lighting conditions, images, and shadow types, leading to a rapid decline in detection accuracy.
4) *DL Method:* This method based on neural networks adopts the convolutional neural network (CNN) model [18], [19], [20] to detect shadows automatically by using supporting training datasets and offers certain advantages.

In recent years, the CNN model [37], [39], [40] has exhibited excellent performance in the field of image processing; however, the CNN-based shadow detection method has not been studied extensively. Shadow detection can be regarded as a semantic segmentation problem. FCN [21], UNet [22], and Deeplab [23] are classic semantic segmentation networks that exhibit good performance in remote-sensing image-processing tasks such as cloud detection. Thus, the use of a semantic segmentation network for shadow detection is worth exploring. Luo et al. [24] proposed the first remote-sensing image shadow detection dataset aerial imagery dataset for shadow detection (AISD) and the corresponding deeply supervised CNN (DSSDNet) for shadow detection. Jin et al. [25] proposed a network based on the global–spatial–context attention module (GSCA-UNet). Luo et al. [26] proposed the edge perception spatial pyramid network (ESPFNet), combining the edge detection method with the neural network method, focusing on the edge features of shadows. Zhu et al. [27] proposed the contextual detail-aware network (CDANet). However, these existing DL methods for shadow detection ignore the similarity between local dark areas and shadows in remote-sensing images and do not design modules specifically for shadow features in terms of spatial relationships, so their ability to extract shadow features is limited and they are prone to incorrectly detect similar but nonshadowed areas as shadows, the details of which are described in Section II-A.

Based on the above observations, to solve the problem of insufficient shadow feature extraction and to further improve the detection accuracy, in this study, we proposed a deep neural network based on multiresolution parallel fusion (MRPF) decoding and the fusion of cross-spatial and channel attention for shadow detection in HR remote-sensing images. The main contributions of this study are as follows.

1) Aiming at the problem that local dark areas in remote-sensing images are easy to be misjudged as shadows, we designed a simplified cross-spatial attention module that can extract the row and column features of feature map pixels and focus on global semantic information.
2) We proposed the channel self-attention module to extract color information from multichannel feature maps, focusing on channels that match shadow color features.
3) MRPF decoding connects all resolutions in parallel to realize HR representation and avoid information loss caused by the process of converting a low-resolution (LR) image into an HR image. In addition, the repeated multiscale fusion between adjacent resolutions enables each resolution representation to repeatedly receive information from other resolutions, which helps improve prediction accuracy.

II. Methodology

In this section, the simplified cross-spatial attention module designed for shadow features and the channel self-attention module designed for multichannel feature maps are discussed. The network is a codec architecture; the encoder uses ResNeXt50 as the backbone, whereas the decoder employs MRPF decoding. All resolutions are connected in parallel, and repeated multiscale fusion is performed between adjacent resolutions to improve the prediction accuracy. The above-mentioned attention modules are added to the MRPF decoder to realize MRPF attention (MRPFA) decoding.

A. Simplified Cross-Spatial Attention

Shadow detection is the process of determining whether the pixels in an image belong to a shadow. However, this is challenging because shadows often have the characteristics of being dark in color and having a regular shape and may sometimes be confused with other ground objects that share these features.

As shown in Fig. 1 (left), area B is the cement pavement of a parking lot, which is dark in color and can easily be mistaken for a shadow compared to the lighter-colored pavement in area A. However, when compared to the actual shadow of

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Fig. 1. Diagram of the shadow.

buildings in area C, it becomes apparent that area B is not a shadow.

In Fig. 1 (right), the red-framed ground object in the remote-sensing image is a roof with a dark color and a regular shape. If viewed locally and when only the dark roof is considered, it can be easily misidentified as a shadow. However, by examining the entire image and comparing the dark roofs with their horizontal and vertical pixels, the true shadow can be identified, and the roof area can be determined to be not a shadow.

Therefore, judging whether a certain area is a shadow should be based not only on the feature information of the area, but also on contextual semantic information. The shadow can be judged only from a global perspective.

Targeting the aforementioned problems, we designed a simplified cross-spatial attention module that references self-attention [34], [35], [38] and CCNet [28] to extract the pixel features of the same row and column as the target pixels in the feature map.

In this module, the input feature map with shape $R^{C\times H\times W}$ is processed by the branches Q and W to realize pixel feature extraction. The process can be expressed as follows:

$$\begin{cases} Q_i = R_i^1(x), & Q_i \in R_i^{C'\times 1\times 1} \\ K_i = R_i^2(x), & K_i \in R_i^{C'\times (H+W-1)\times 1} \end{cases} \tag{1}$$

where Q_i denotes the feature of the ith pixel extracted from the input image by the first branch; $R_i^1(x)$ denotes the 1-D feature of the ith pixel extracted from the input image x by the first branch; $R_i^{C'\times 1\times 1}$ denotes the feature of the image with the number of channels outputted by the first branch being C', the number of pixel rows being one and the number of pixel columns being one; K_i denotes the feature of the ith pixel extracted from the input image by the second branch; $R_i^2(x)$ represents the feature of the pixel in the same row as the ith pixel extracted from the input image x; H is the total number of all pixel rows in the input image; W is the total number of all pixel columns in the input image; and $R_i^{C'\times (H+W-1)\times 1}$ represents the feature of the image with the number of channels outputted by the second branch being C', the number of pixel rows being $(H+W+1)$, and the number of pixel columns being one. C' is usually set as one-eighth of the number of channels C in the input feature map.

Next, the Key vector is reshaped and cross-multiplied with the query vector, and the vector D with the shape of $(H, W, H+W-1, 1)$ is obtained.

This process can be mathematically expressed as follows:

$$d_{i,j} = K_{i,j}^T Q_i \tag{2}$$

where $i = 1, 2, \ldots, h\times w$ represents the spatial location index of a pixel located in a row and a column represents the features of the same row and column extracted using the crisscross method, $Q_i \in R^{(C/8)\times 1}$, $d_{i,j} \in R^{(h+w-1)\times 1}$, and $d_{i,j} \in R^{(h+w-1)\times 1}$.

Vector D is normalized using the softmax function to obtain vector A. When A is copied, transposed, and multiplied by itself, the attention weight of each point in the feature map is obtained as follows:

$$\text{out}_{i,j} = A_{i,j}^T A_{i,j} \tag{3}$$

where $A_i \in R^{(h+w-1)\times 1}$, $A_i^T \in R^{1\times (h+w-1)}$, and $\text{out}_{i,j} \in R^{h\times w\times 1\times 1}$.

The dimensions of $\text{out}_{i,j}$ are extruded and exchanged to obtain the shape of $(1, H, W)$, which is the feature map with horizontal and vertical attention applied, and it is multiplied with the original map to realize criss-cross attention. The structure of the simplified cross-spatial attention module is shown in Fig. 2.

This module can fuse the horizontal and vertical information of each pixel in the feature map to the current point so that each position point can concentrate the crisscross direction features. This aids in shadow detection because the module focuses on the contextual semantic information of an area to determine shadow areas, enhancing the ability to extract spatial information from the feature map and enabling more accurate determination of small and suspicious shadow regions. In addition, both extract single-point crisscross direction features, CCNet utilizes three branches, whereas the proposed module requires only two branches. Thus, the proposed approach yields good performance unchanged while reducing matrix computation complexity.

B. Channel Self-Attention Module

There are many multichannel feature maps in the network, some of which can highlight shaded features, while others focus on highlighting features in nonshaded regions. For shadow detection networks, feature learning should be more biased toward the channels of image shadow features. In this section, we designed the channel self-attention module to filter these feature channels and highlight features in the shadow regions, which facilitates the shadow detection task.

In the module, the input feature map with shape $R^{C\times H\times W}$ is processed by parallel branches Q and K. After 1×1 convolution in branch Q, the channel dimension is changed to 1 to obtain the Query vector and reshaped into $R^{(H*W)\times 1}$. The K branch is used to obtain the Key vector and reshaped into $R^{C\times (H*W)}$. Next, the outputs of the branches are multiplied to obtain the attention of the channel direction; the shape of the output is $R^{C\times 1\times 1}$. Next, the softmax function is used, and the result is passed through two fully connected layers,

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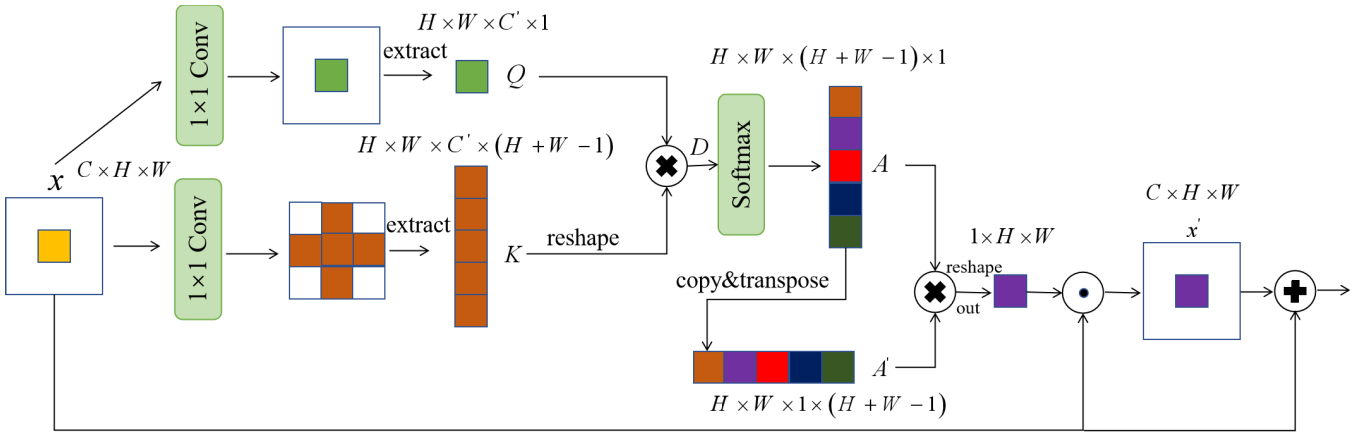


Fig. 2. Simplified cross-spatial attention module.

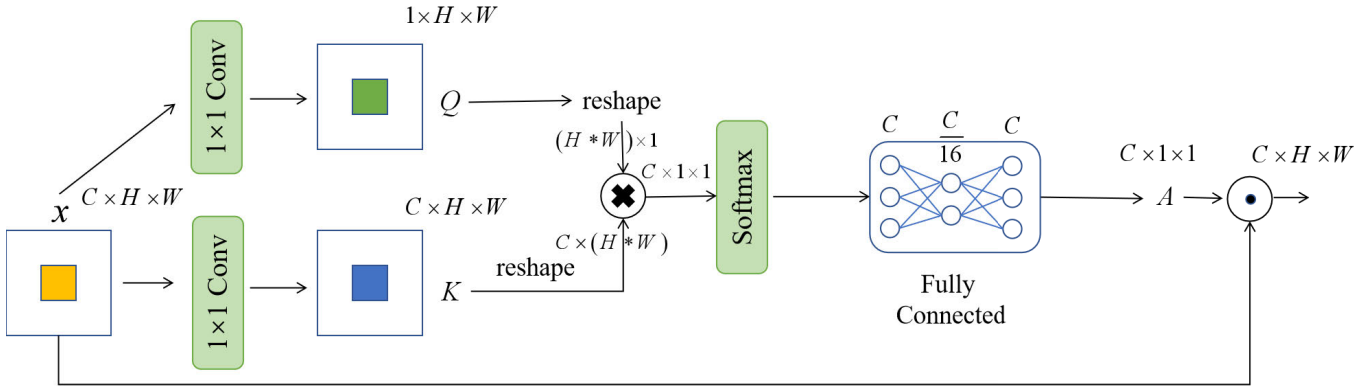


Fig. 3. Channel self-attention module.

in which the number of neurons is reduced to 1/16 of the original number first and then restored to the original second. Finally, channel self-attention is realized by multiplying the channel attention and the input feature map. This process can be mathematically expressed as follows:

$$\text{out} = \text{Scale}(x, \text{FC}(\sigma(f(x)))) \tag{4}$$

where X represents the input feature map, f refers to the processing and fusion of branches Q and K, σ denotes the softmax function, FC denotes the two-layer full connection operation, and $\text{Scale}(\cdot, \cdot)$ refers to the multiplication operation. The structure of the channel self-attention module is shown in Fig. 3.

The designed module can extract color information from feature maps, pay greater attention to channels that match shadow color characteristics, overcome the problem of limited extraction of shadow features in existing technologies, reduce missed and erroneous detection of shadow regions, and improve shadow detection accuracy.

C. Network Structure

The overall network structure proposed in this article is shown in Fig. 4(a). The network is a codec architecture, ResNeXt50 [29] is employed as the backbone of the encoder and outputs four resolutions: $R^{64\times128^2}$, $R^{256\times64^2}$, $R^{512\times32^2}$, and $R^{2048\times16^2}$. HRNet [30], [31] is employed as the decoder for realizing the high-resolution (HR) representation and repeated fusion of features. We designed a multiresolution parallel fusion (MRPF) decoder, which has two characteristics: parallel and fusion. First, the four resolutions from the encoder output are connected in parallel to maintain HR representation, avoid information loss caused by the process of obtaining HR representation from the LR representation and make the feature map more accurate in space. In addition, repeated multiscale fusion is performed between two adjacent resolutions to enable each resolution representation to repeatedly receive information from other resolutions and improve the HR representation obtained from the LR representation of the same depth and similar level, which helps improve the prediction accuracy. The decoder comprises three MRPF decoding blocks. Each decoding block has two resolutions, which are obtained by upsampling, splicing, and fusion of the resolution from the encoder output. Next, the repeated fusion of the two basic resolution features is performed.

In the MRPF block [see Fig. 4(b)], conversion from the LR representation to the HR representation is realized through 1×1 convolution and upsampling. The new LR representation is obtained through two steps. First, the LR representation is obtained using the cascaded simplified cross-spatial attention and channel self-attention module. Next, the HR representation is obtained through 3×3 convolution downsampling with step 2. The new HR representation is obtained using three steps. First, the original HR is used. Next,

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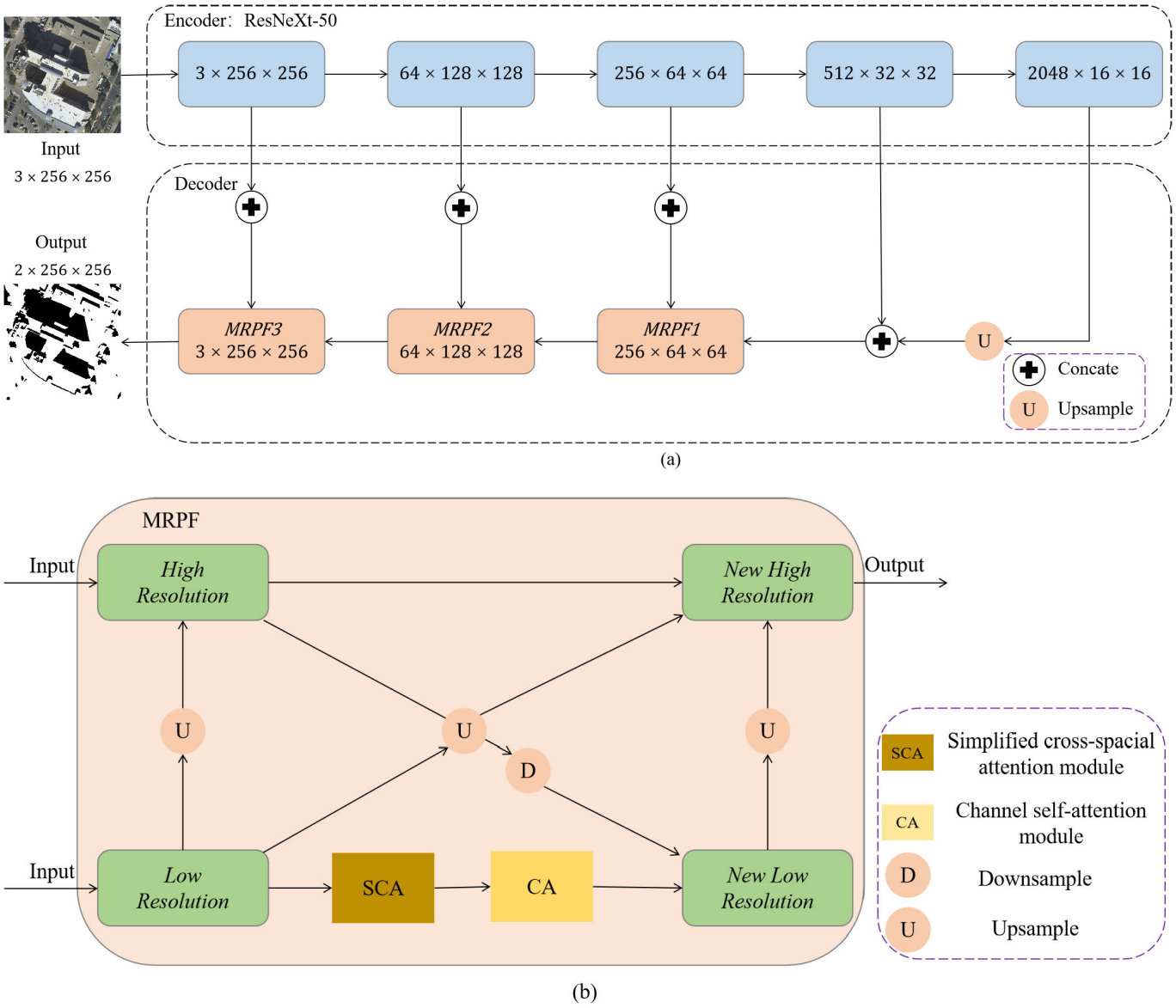


Fig. 4. (a) Overall network structure proposed in this article. (b) MRPF block.

the LR representation is obtained using 1 × 1 convolution and upsampling. Third, new LR representation is obtained through 1 × 1 convolution and upsampling.

Through MRPF decoding, the extraction of shadow features can be improved, and the accuracy of shadow detection in remote-sensing images can be improved.

III. Experiments and Results

In this section, first, we present the dataset and its augmentation methods, as well as the experimental setup details, such as the loss function. Next, we perform a quantitative comparison with other algorithms through experiments and conduct the ablation experiment to demonstrate the effectiveness of each designed module. Finally, the experimental results are visualized and analyzed.

A. Datasets and Implementation Details

1) Experimental Dataset: The AISD [24] remote-sensing shadow dataset from Wuhan University and the shadow semantic annotation database (SSAD) [32] shadow dataset from Lanzhou University were used in this study.

AISD has 514 images and corresponding labels. It contains aerial remote-sensing shadows of buildings and other features in Austin, Chicago, Innsbruck, Tyrol, and Vienna. Representative shadow images and their corresponding labels of the datasets of five different regions are shown in Fig. 5.

As can be seen in Image 1, the buildings in Austin are relatively scattered, and there are many vacant spaces between the buildings. Chicago is a commercial city with tall and dense buildings and many skyscrapers. Therefore, when the oblique sunlight is blocked by high-rise buildings, shadows are cast on the ground, as can be seen in Image 2. The buildings in Innsbruck are low but dense (Image 3). In Tirol, buildings are low and sparse, and trees are widely distributed on the site outside the buildings (Image 4). The buildings in Vienna have a closed-ring structure, with trees and other vegetation planted inside (Image 5). The surface features of these five cities have different structural characteristics, making the dataset rich, diverse, and comprehensive, which is helpful for training.

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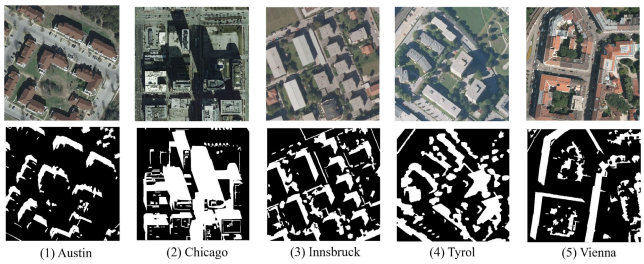


Fig. 5. AISD dataset pictures and their labels.

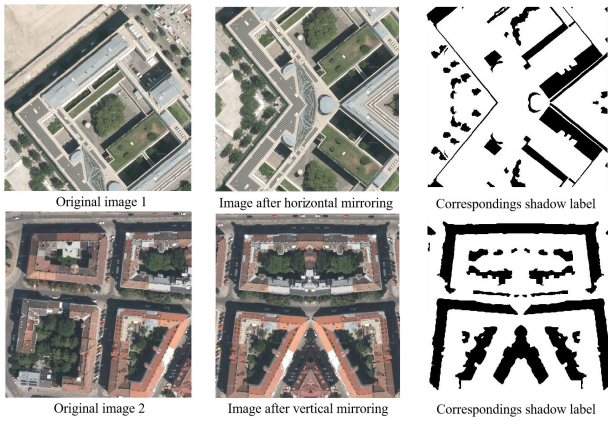


Fig. 6. Data enhancement diagram.

The shadow sources in the AISD dataset mainly include shadows cast by buildings and vegetation. The shadows cast by buildings are regular in shape and distributed in blocks, whereas the shadows cast by vegetation differ in shape and are distributed in an irregularly scattered form. The shadow area of buildings is more than that of vegetation. Therefore, the remote-sensing shadow detection model should perform shadow detection of regular, blocky, large-area buildings as well as that of irregular, scattered, small-area vegetation. Which is challenging.

We divided the dataset into the training verification set and the test set in the ratio of 9:1. Thus, the number of images in the test set was 51. However, the training set contained only 463 images with sizes ranging from 256 × 256 to 1692 × 1692 pixels, which is not sufficient. Therefore, it is necessary to enhance the dataset and increase the sample size.

To enhance the dataset, we uniformly cropped the images into small patches with a size of 256 × 256 pixels. For cropping, a 256 × 256 window was placed on the original image and the appropriate area was selected. The objective was to retain sufficient shadow pixels in the figure. Mirror transformation was performed randomly along the horizontal or vertical directions, as shown in Fig. 6.

The image after transformation retained the characteristics of the building and shadow shape rules, and the contents of the image and label were still recognizable after the enhancement. By cropping images into patches, the original size of only 463 training verification set images was expanded to 5523, thus greatly increasing the sample size. Furthermore, by dividing the training set and verification set in the ratio of 8:2, 4418 training sets and 1105 verification sets were obtained.

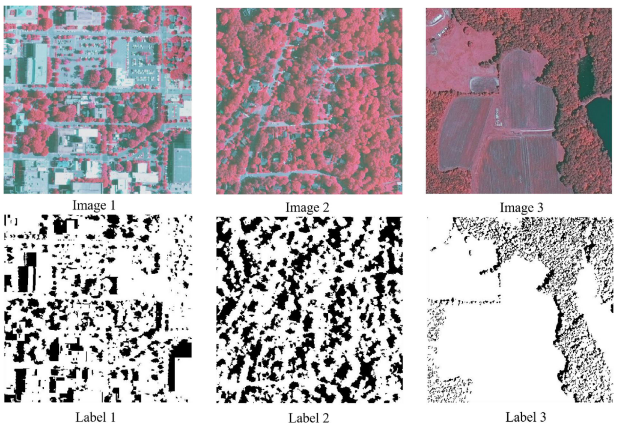


Fig. 7. SSAD dataset pictures and their labels.

The second dataset used is SSAD, which has 103 pairs of images with shadows and their shadow labels, covering urban, suburban, and rural areas, with six shadow types, namely buildings, trees, grass, roads, water, and farmland. Each image is 500 × 500 pixels in size. Representative shadow images of this dataset and their corresponding labels are shown in Fig. 7.

The shadow distribution of urban areas is shown in Image 1. The shadows in urban areas are mainly those cast by buildings, but also contain some shadows cast by vegetation, roads, and other structures. The shadow distributions of suburban and rural areas are shown in Images 2 and 3, respectively. The shadows in the suburban and rural areas are dominated by vegetation. Compared with the AISD dataset, the SSAD dataset has richer shadow types and higher requirements for the extraction ability of various shadow features.

We divided the SSAD dataset into the training verification set and the test set in the ratio of 7:3. Accordingly, 82 training verification sets and 21 test sets were obtained. By cropping the images of datasets into small patches with a size of 256 × 256 pixels, the training verification set was expanded to 2788 picture pairs. Furthermore, by dividing the training set and verification set in the ratio of 9:1, 2509 training sets and 279 verification sets were obtained.

2) Loss Function and Training Details: Cross-entropy (CE) loss and dice loss were used as loss functions to guide network training. CE loss can be expressed as follows:

$$L_1 = -\big[y \log(p) + (1-y)\log(1-p)\big] \tag{5}$$

where y represents the label of the sample: a positive class is assigned a value of 1, whereas a negative class is assigned a value of 0. p is the probability that the sample is predicted to be positive.

The expression for Dice loss is

$$L_2 = 1 - \frac{2^{*}\mathrm{TP}}{\mathrm{FP} + 2^{*}\mathrm{TP} + \mathrm{FN}} \tag{6}$$

where TP, FP, and FN represent the statistical values of true cases, false-positive (FP) cases, and false-negative (FN) cases, respectively. The use of dice loss helps balance positive and

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negative samples. The final loss value is the sum of the two losses.

In the training process, a gradient descent algorithm was used to optimize the model parameters, and the Adam optimizer was used. A total of 100 epochs were trained, and the initial learning rate was set as 1e-4 for the first 30 epochs, 1e-5 for 30–60 epochs, 5e-6 for 60–80 epochs, and 1e-6 for 80–100 epochs. For parameter initialization, the ResNeXt50 backbone network uses pretrained parameters to obtain better training results. The backbone network parameters are frozen in the first 60 epochs and unfrozen in the last 40 epochs.

Details regarding the experimental platform are as follows: Ubuntu-16.04.1 LTS operating system, CUDA-10.2 environment. The training was performed using two NVIDIA GeForce GTX 2080Ti GPUs. The batch size was set as 4. The training time of the proposed network model is approximately 5 h.

3) Evaluation Metrics: To evaluate the effectiveness of the proposed method, the commonly used evaluation indexes were selected: F_1 score, overall accuracy (OA), intersection over union (IOU), and bit error rate (BER). For the former three, a higher value indicates better performance. In contrast, for BER, the lower the value, the better. Their expressions are as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad F_1 = \frac{2^*P^*R}{P+R} \tag{7}$$

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}+\mathrm{TN}} \tag{8}$$

$$\mathrm{IOU} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \tag{9}$$

$$\mathrm{BER} = 1 - \frac{1}{2}\left(\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} + \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}}\right) \tag{10}$$

where true positive (TP) means both the real situation and the predicted result are shadows, true negative (TN) means only the real situation is a shadow, but the result is not predicted as a shadow, FP means the real situation is not a shadow, but the result is predicted as a shadow, and FN means both the real situation and the predicted result are not shadows.

B. Ablation Study

In this section, the ablation study was conducted to demonstrate the role played by the network and modules proposed in this article. To better control the parameters for comparison, the ablation study was conducted by modifying the network and adding modules step by step based on the encoder–decoder network structure, Experiment 1: encoder–decoder network; Experiment 2: encoder–decoder network with ResNeXt50 as the backbone; Experiment 3: replacing the decoder with MRPF decoding, while the decoding block does not contain the attention module; Experiment 4: adding the simplified cross-spatial attention module to the decoding block; Experiment 5: further add the channel self-attention module to the decoding block. The five experiments are conducted under the same training conditions.

As can be seen in Table I, it can be seen that with the addition of the designed decoding part and attention modules, the F_1 score, OA, BER, and IOU improved in the experimental results on the AISD dataset. Among them, the addition of MRPF decoding increases the F_1 score by 4%, and the addition of the simplified cross-spatial attention module and the channel self-attention module makes the F_1 score finally reach 92.6%.

TABLE I
ABLATION STUDY ON AISD

Method	Evaluation			
	F1 score(%)	OA(%)	BER(%)	IOU(%)
encoder–decoder network	87.77	94.66	8.02	78.69
+ResNeXt50 Backbone	88.28	94.87	7.50	79.51
+multi resolution parallel fusion decoding	92.22	95.95	5.49	85.56
+Simplified cross-spacial attention	92.42	96.03	5.41	85.9
+Channel self-attention	92.60	96.11	5.42	86.24

C. Experimental Comparison by Using AISD

1) Comparison of Algorithms: To verify the performance of the proposed method, we performed experiments on the AISD dataset by using DLA-PSO [33], Deeplabv3+, CGAN [36], MTNT [42], DSSDNet, GSCA-UNet, CAD-Net [43], ESPFNet, FSDNet [41], and CDANet for comparison. Among them, DLA-PSO is the traditional shadow detection method, Deeplabv3+ and CGAN are DL-based SOTA methods, while the other seven are recently published SOTA methods in the field of shadow detection.

To ensure the fairness of the experiment, we set the backbone networks of Deeplabv3+, CGAN, FSDNet, and GSCA-UNet in the comparison experiments to also be ResNeXt50, with pretrained parameters, adopting exactly the same training strategy as the proposed method.

Note that the other methods do not disclose their source code and are difficult to reproduce. Therefore, here we directly extract the experimental results of their papers and use the missing values to represent the unrepresented parameters.

As can be seen in Table II, the proposed network exhibited better performance in terms of F_1 score. In addition, compared with GSCA-UNet, except for OA, the proposed network exhibited better performance in terms of BER and IOU, with the IOU higher by more than one percentage point, thus demonstrating the effectiveness and superior performance of the proposed algorithm.

D. Experimental Comparison by Using SSAD

1) Comparison of Algorithm: To further test the performance of the proposed model, the model was trained and tested on the SSAD dataset, and FCN, UNet, Deeplabv3+, RSAM [32], MTNT, DSSDNet, FSDNet, and CDANet were selected for comparison. Among them, FCN, UNet, and Deeplabv3+ are DL-based SOTA methods, while the other four are recently published SOTA methods in the field of

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TABLE II
COMPARISON OF SOTA METHODS ON AISD

Method	Evaluation			
	F1 score(%)	OA(%)	BER(%)	IOU(%)
Deeplabv3+ (2018)	91.15	95.46	5.66	83.74
CGAN (2018)	92.09	95.61	**5.35**	85.35
MTNT (2020)	90.68	-	-	-
DSSDNet (2020)	91.79	95.57	6.24	-
GSCA-UNet (2020)	91.69	**96.29**	5.51	84.88
CADNet (2021)	91.21	-	-	-
ESPFNet (2021)	92.04	-	-	-
FSDNet (2021)	92.36	95.81	5.53	85.54
DLA-PSO (2022)	88.73	94.9	-	-
CDANet (2022)	92.41	-	-	-
Proposed Method	**92.60**	96.11	5.42	**86.24**

TABLE III
COMPARISON OF SOTA METHODS ON SSAD

Method	Evaluation			
	F1 score(%)	OA(%)	BER(%)	IOU(%)
FCN (2015)	73.51	88.63	25.57	62.18
Unet (2015)	67.77	81.10	34.72	55.07
Deeplabv3+ (2018)	81.31	92.13	16.7	70.96
RSAM (2020)	-	90.6	-	-
MTNT (2020)	79.02	-	-	-
DSSDNet (2020)	79.31	-	-	-
FSDNet(2021)	82.65	92.25	15.34	73.71
CDANet (2022)	83.96	-	-	-
Proposed Method	**85.21**	**93.38**	**14.99**	**75.84**

shadow detection. The same model training method was employed for all the algorithms.

To ensure the fairness of the experiment, we set the backbone networks of Deeplabv3+ and FSDNet in the comparison experiments to also be ResNeXt50, with pretrained parameters. The above networks, as well as FCN and UNet, which both use the exact same training strategy as the proposed method.

Note that the other methods do not disclose their source code and are difficult to reproduce. Therefore, here we directly extract the experimental results from their papers or related papers and use the missing values to represent the unrepresented parameters.

As can be seen from Table III, the proposed method exhibited the best results in terms of every index, leading other algorithms by 2\% points in some metrics.

2) Comparison of Runtime and Number of Parameters: The runtime and number of parameters of the proposed model were compared with those of other advanced algorithms in Table IV.

TABLE IV
COMPARISON OF RUNTIME AND PARAMETER

Method	Evaluation	
	Time(s)	Param(M)
FCN (2015)	0.250	512.2
Unet (2015)	0.248	120.61
Deeplabv3+ (2018)	0.093	226.36
MTNT (2020)	0.145	169.21
FSDNet(2021)	0.382	186.85
CDANet (2022)	0.171	164.23
Proposed Method	0.265	134.37

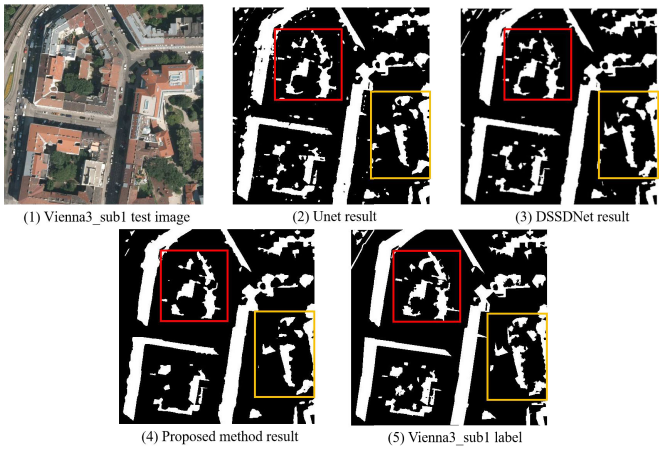


Fig. 8. Shadow detection results from the AISD dataset Vienna.

And the testing experiments of each model were conducted on two NVIDIA GeForce GTX 2080Ti GPUs. Compared with FCN, UNet, Deeplabv3+, MTNT, FSDNet, and CDANet, the proposed model requires fewer parameters and yields superior performance. Compared with UNet, the proposed model requires a slightly larger number of parameters but yields superior performance. Although the proposed model runs slightly longer than some other algorithms, its accuracy in shadow detection is optimal.

E. Qualitative Analysis

Qualitative analysis was performed by comparing the shadow pixel prediction results of different models. First, for the AISD dataset, UNet and DSSDNet were selected as references for comparison with the proposed method. The prediction results of the models are as follows.

In Fig. 8, Images 1 and 5 are test images and the corresponding shadow labels of the city of Vienna, respectively. Image 2 is the prediction result after training using the UNet model. The training method adopted was consistent with the proposed method, and the results added many scattered shadows of false detection. Dark vehicles on some streets and dark skylights on the roofs of buildings were misjudged as shadows, and some areas of white vehicles in the shadow coverage area of buildings were misjudged as not shadows. Image 3 is

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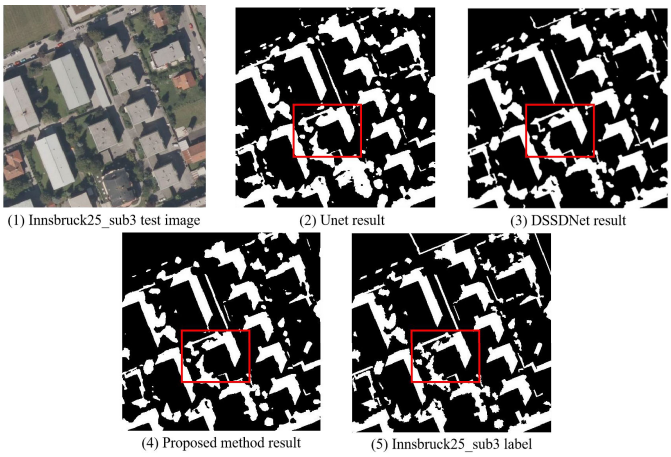


Fig. 9. Shadow detection results from the AISD dataset Innsbruck.

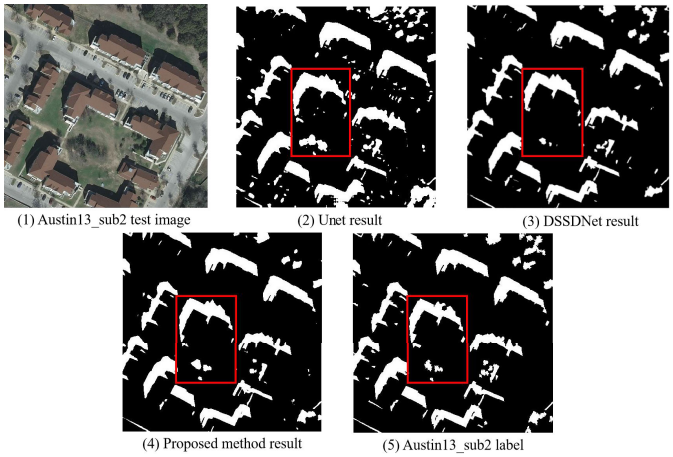


Fig. 10. Shadow detection results from the AISD dataset Austin.

the prediction result obtained using DSSDNet; vehicles were misjudged as shadows, and some dot-like blocks that were mistakenly identified as shadows were observed. Image 4 is the prediction result obtained using the proposed method; most of the shadows of buildings and vegetation were correctly detected, which is closest to the actual label.

The shadow detection results for Innsbruck on the test dataset are shown in Fig. 9. The resolution of remote-sensing Image 1 in Fig. 8 is 496 × 496; the buildings in Austin were scattered, and there were many vacant spaces between buildings. Image 5 is the corresponding shadow label and contains mainly building shadows and a small number of vegetation shadows. Furthermore, as can be seen from the prediction result obtained using UNet (Image 2), the result was chaotic; dark glasses of dark vehicles or light vehicles were misjudged as shadows, and the detection of vegetation shadows was highly inaccurate. As can be seen from the prediction result obtained using DSSDNet (Image 3), some branches and vehicles were incorrectly judged as shadows. Image 4 is the prediction result obtained using the proposed method; the judgment of building shadows was accurate, but some vegetation shadows in the upper right corner could not be detected.

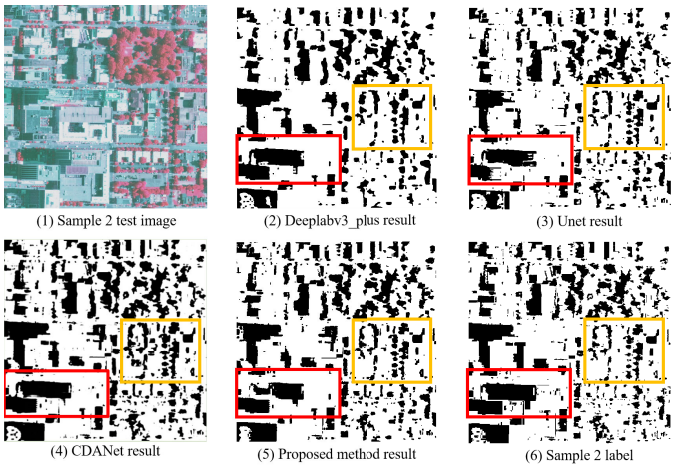


Fig. 11. Shadow detection results from the SSAD dataset.

Fig. 10 shows the shadow detection results of Austin on the test dataset. The resolution of remote-sensing Image 1; buildings in Innsbruck are low but dense. Image 5 is the corresponding shadow label. The shadow of buildings was blocky, and the shadow of vegetation was irregular. As can be seen from the prediction result obtained using UNet (Image 2), the result was not precise; the detection of vegetation shadows was often more or less than the actual label. As can be seen from the prediction result obtained using DSSDNet (Image 3), some strip shadows at the edge of roads could not be detected. Finally, as can be seen from the prediction result obtained using the proposed method (Image 4), the overall detection accuracy was high.

For the SSAD dataset, Deeplabv3+, UNet, and CDANet were used as a reference for comparison with the proposed method. The prediction results obtained using the models are as follows.

Fig. 11 shows the shadow detection results on the test dataset. As can be seen from the prediction result obtained using Deeplab v3+(Image 2), shadow details were not detected accurately. As can be seen from the prediction result obtained using UNet (Image 3), the detection of shadows of buildings and vegetation was inaccurate. Image 4 is the prediction result obtained using CDANet; more shadow regions than the actual shadow regions were detected. Finally, as can be seen from the prediction result obtained using the proposed method (Image 5), most shadows of buildings and vegetation were correctly detected, which is closest to the real label.

IV. Conclusion

We developed a deep neural network model for shadow detection in HR remote-sensing images. First, the shadow characteristics were analyzed, and it was found that the shadow and some disturbed ground objects have dark color characteristics, and the right judgment can be made only by combining the global information. Thus, we developed the simplified cross-spatial attention module to extract row and column features. In addition, we developed the channel self-attention module to meet the channel information requirement of the multichannel feature map in a network. The proposed network

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is a codec architecture; ResNeXt50 is used as the backbone of the encoder, whereas an MRPF decoder is used as the decoder. All resolutions are connected in parallel, and repeated multiscale fusion is performed between adjacent resolutions to improve the prediction accuracy. We added these attention modules into the MRPF decoder to realize MRPFA decoding. Finally, we used AISD and SSAD shadow datasets for model training, verified the effectiveness of the developed modules by conducting an ablation study, and demonstrated the superior performance of the proposed network by performing a comparative analysis with existing advanced algorithms.

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Jing Zhang received the B.Sc. degree in information engineering and the Ph.D. degree in information and communication engineering from Xi'an Jiaotong University, Xi'an, China, in 2003 and 2009, respectively.

From September 2007 to September 2008, she was a Visiting Ph.D. Student with Mississippi State University, Starkville, MS, USA. She is currently with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an. Her research interests include image processing and the design of unmanned systems.



Xinlong Shi is currently pursuing the master's degree with Xidian University, Xi'an, China.

His research interests include computer vision, remote sensing, and deep learning.



Congyao Zheng is currently pursuing the master's degree with Xidian University, Xi'an, China.

Her research interests include computer vision, remote sensing, and image compression algorithm.



Jun Wu is currently pursuing the master's degree with Xidian University, Xi'an, China.

His research interests include computer vision, remote sensing, and machine learning.



Yunsong Li (Member, IEEE) received the bachelor's degree in image transmission and processing, the master's degree in communication and information systems, and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1996, 1999, and 2002, respectively.

In 2009, he was selected into the New Century Excellent Talents Program of the Ministry of Education. He is currently the Deputy Dean at the Aerospace Research Institute, Xidian University, an Academic Leader of Communication and Information Systems, the Doctoral Supervisor of Communication and Information Systems, and the Standing Director of Shaanxi Provincial Graphics and Image Society.

Dr. Li is a member of the Scientific Application Expert Committee of Lunar Exploration Engineering, and the Deep Space Exploration Technology Professional Committee of the Chinese Astronautical Society.

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